



University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Second Year Examination - Semester II - UCSC AY20 [held in March/ April 2024]

SCS2213 – Electronics and Physical Computing

(Two (2) Hours)

Answer ALL questions

Number of Pages = 13

Number of Questions = 50 MCQ



200

To be completed by the candidate

Index Number:

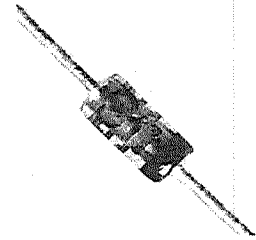
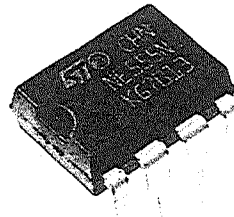
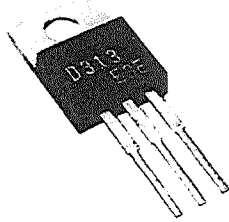
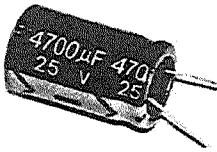
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Important Instructions to candidates:

- I. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- II. This paper consists of **50 Multiple-Choice Questions (MCQs) in 13 pages** (including the Cover Page).
- III. Each MCQ question will have 5 (five) choices with **only one correct/best answer**. Selecting more than one option per question will result in zero marks being awarded.
- IV. Each MCQ question will carry equal marks.
- V. Answer ALL questions and submit your answers in the **MCQ Answer Sheet** provided. Please read the instructions given in the back side of the MCQ answer sheet as well.
- VI. Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.

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Consider the following electronic components to answer the questions 1-2.



1) Which option accurately identifies the components in the diagram from left to right?

- (a) Battery, Triac, IC, and Zener Diode
- (b) Capacitor, Voltage Regulator, IC, and Zener Diode
- (c) Capacitor, Triac, Optocoupler, and Zener Diode
- (d) Electrolytic Capacitor, Transistor, IC, and Diode
- (e) Battery, Voltage Regulator, Microcontroller, and Diode

2) Which of the following is not a device number/value?

- (a) D313
- (b) KG7013
- (c) 4700uF
- (d) 1N4148
- (e) NE555N

3) What is the relationship between voltage, current, and resistance in a DC circuit according to Ohm's Law?

- (a) $I = VR$
- (b) $R = IV$
- (c) $V = IR$
- (d) $V = I^2R$
- (e) $R = I/V$

4) Given a resistor with a resistance of 120 ohms and a current of 0.5 amperes flowing through it, what is the voltage across the resistor?

- (a) 12 volts
- (b) 0.0042 volts
- (c) 240 volts
- (d) 60 volts
- (e) 0.0006 volts

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5) In a series circuit with three resistors of 2 ohms each, what is the total resistance?

- (a) 2 ohms
- (b) 4 ohms
- (c) 6 ohms
- (d) 8 ohms
- (e) 10 ohms

6) When capacitors are connected in series, what happens to the total capacitance?

- (a) It decreases
- (b) It increases
- (c) It remains the same
- (d) It becomes zero
- (e) It becomes infinite

7) What is the resistance value of a resistor with the color bands: Brown, Black, Orange, Gold?

- (a) 100 ohms
- (b) 10 ohms
- (c) 1000 ohms
- (d) 10 kohms
- (e) 100 kohms

8) If a yellow LED with a forward voltage drop of 2.1 volts is connected to a 12-volt battery, and a 330-ohm resistor is used to limit the current, what is the forward current through the LED?

- (a) 45 mA
- (b) 40 mA
- (c) 35 mA
- (d) 30 mA
- (e) 25 mA

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- 9) If an LED has a forward voltage drop of 2.2 volts and is connected to a 9-volt battery, how many LEDs with the same properties can be connected in series without exceeding the battery voltage?

(a) 3 LEDs
(b) 4 LEDs
(c) 5 LEDs
(d) 6 LEDs
(e) 2 LEDs

- 10) You have four LEDs, each with a forward voltage drop of 2.5 volts, and a 12-volt battery. If you connect them in parallel with a single resistor to limit the current to 20mA per LED, what resistor value should you use?

(a) 120 ohms
(b) 125 ohms
(c) 250 ohms
(d) 32 ohms
(e) 475 ohms

- 11) If a battery supplies a current of 5 amps to a circuit for 2 minutes, how much charge has passed through the circuit?

(a) 300 C
(b) 600 C
(c) 120 C
(d) 60 C
(e) 10 C

- 12) What is the power consumed by a device if a voltage of 12 volts is applied across it and a current of 2 amps flows through it?

(a) 6 W
(b) 12 W
(c) 24 W
(d) 14 W
(e) 20 W

13) Which of the following accurately describes the characteristic curve of a diode?

- (a) No relationship between voltage and current
- (b) Linear relationship between voltage and current
- (c) Inverse relationship between voltage and current
- (d) Quadratic relationship between voltage and current
- (e) Exponential relationship between voltage and current

14) Which type of biasing is typically used to ensure a diode is conducting current?

- (a) Reverse biasing
- (b) Zero biasing
- (c) AC biasing
- (d) Forward biasing
- (e) Common biasing

15) What is the primary function of a transistor in an electronic circuit?

- (a) Convert electrical energy to mechanical energy
- (b) Control current flow
- (c) Store electrical charge
- (d) Generate electromagnetic waves
- (e) Act as a voltage source

16) Which type of transistor has a thin base region sandwiched between emitter and collector regions?

- (a) Bipolar Junction Transistor (BJT)
- (b) Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)
- (c) Junction Field-Effect Transistor (JFET)
- (d) Insulated-Gate Bipolar Transistor (IGBT)
- (e) Schottky Transistor

17) In a common-emitter configuration of a transistor, where is the input signal applied?

- (a) Emitter
- (b) Base
- (c) Collector
- (d) Drain
- (e) Source

- 18) Which type of diode is commonly used as a voltage reference or voltage regulator?
- (a) Schottky diode
 - (b) Tunnel diode
 - (c) Zener diode
 - (d) Light-emitting diode (LED)
 - (e) Photodiode
- 19) What is the common-emitter current gain (β) of a transistor if the collector current (I_c) is 4 mA and the base current (I_b) is 40 μ A?
- (a) 50
 - (b) 75
 - (c) 100
 - (d) 125
 - (e) 150
- 20) In a common-emitter configuration, a transistor with a base current (I_b) of 40 μ A and a current gain (β) of 150 is connected to a 500-ohm resistor in the collector circuit. What is the voltage drop across the resistor if the collector current (I_c) is 6 mA?
- (a) 5.4 V
 - (b) 4.5 V
 - (c) 3.6 V
 - (d) 3 V
 - (e) 2.4 V
- 21) What type of switch is commonly used in electronic devices for momentary contact applications such as doorbells?
- (a) Toggle switch
 - (b) Rocker switch
 - (c) Push-button switch
 - (d) Slide switch
 - (e) Rotary switch

22) What type of switch is commonly used in electrical circuits where multiple positions are required, such as selecting different modes or speeds?

- (a) Toggle switch
- (b) Rocker switch
- (c) Push-button switch
- (d) Slide switch
- (e) Rotary switch

23) What is the output of an XOR gate if both of its inputs are 1?

- (a) 0
- (b) 1
- (c) Depends on the previous state
- (d) Undefined
- (e) None of the above

24) In a 2-input multiplexer (MUX), how many select lines are needed to choose between the input channels?

- (a) 1
- (b) 2
- (c) 3
- (d) 4
- (e) 5

25) Which type of flip-flop changes its state at the rising edge of the clock signal?

- (a) T flip-flop
- (b) JK flip-flop
- (c) D flip-flop
- (d) SR flip-flop
- (e) Master-slave flip-flop

26) How many flip-flops are required to store a 4-bit binary number?

- (a) 2
- (b) 3
- (c) 4
- (d) 8
- (e) 16

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- 27) Which type of logic circuit retains memory of its previous state?
- (a) Combinational logic circuit
(b) Sequential logic circuit
(c) Synchronous logic circuit
(d) Asynchronous logic circuit
(e) Multivibrator circuit
- 28) Which type of flip-flop has two inputs and can toggle its output between states?
- (a) D flip-flop
(b) JK flip-flop
(c) T flip-flop
(d) SR flip-flop
(e) Master-slave flip-flop
- 29) Which type of logic circuit produces an output solely based on its current inputs?
- (a) Sequential logic circuit
(b) Synchronous logic circuit
(c) Multivibrator circuit
(d) Asynchronous logic circuit
(e) Combinational logic circuit
- 30) What is the primary function of a 555 timer IC in digital electronics?
- (a) Voltage regulation
(b) Synchronous logic circuit
(c) Multivibrator circuit
(d) Asynchronous logic circuit
(e) Combinational logic circuit
- 31) Which of the following is NOT a common feature of an oscilloscope?
- (a) Triggering
(b) FFT analysis
(c) Probe compensation
(d) Voltage regulation
(e) Timebase adjustment

32) What is the primary function of a digital multimeter (DMM)?

- (a) Store and analyze data
- (b) Test network connectivity
- (c) Generate waveforms
- (d) Measure voltage, current, and resistance
- (e) Amplify signals

33) An oscilloscope is set to display a voltage waveform with a sensitivity of 0.2 V/div. If the displayed waveform spans 6 divisions vertically, what is the peak-to-peak voltage of the signal?

- (a) 1.2 V
- (b) 1.8 V
- (c) 2.4 V
- (d) 3.6 V
- (e) 4.8 V

34) Which of the following is an example of an IoT application?

- (a) Online gaming
- (b) Video streaming
- (c) Social media networking
- (d) Smart home automation
- (e) Web browsing

35) What is the role of a bootloader in a microcontroller?

- (a) Control the operation of input/output devices
- (b) Convert analog signals to digital signals
- (c) Store and retrieve program instructions
- (d) Initialize the microcontroller and load program code into memory
- (e) Interface with external sensors

36) What is the purpose of the **setup()** function in Arduino programming?

- (a) To define variables and constants
- (b) To initialize hardware settings and configurations
- (c) To declare functions
- (d) To execute the main program loop
- (e) To perform mathematical calculations

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37) What is the syntax for declaring an integer variable in Arduino programming?

- (a) variable name = value;
- (b) variable name;
- (c) declare variable name;
- (d) set variable name = value;
- (e) int variable name = value;

38) What is the purpose of PWM in Arduino?

- (a) To generate digital signals
- (b) To generate analog signals
- (c) To perform mathematical calculations
- (d) To communicate with external devices
- (e) To control the speed of motors and brightness of LEDs

39) In PWM, what does the duty cycle represent?

- (a) The frequency of the PWM signal
- (b) The voltage level of the PWM signal
- (c) The duration of the ON state compared to the total period
- (d) The duration of the OFF state compared to the total period
- (e) The amplitude of the PWM signal

40) What does the following Arduino code snippet do?

```
int ledPin = 13;

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  digitalWrite(ledPin, HIGH);
  delay(1000);
  digitalWrite(ledPin, LOW);
  delay(1000);
}
```

- (a) Blinks an LED connected to pin 13 every second
- (b) Reads the state of a button connected to pin 13
- (c) Prints the status of LED to the serial monitor
- (d) Controls the speed of a motor
- (e) None of the above

41) What does the following Arduino code snippet do?

```
int buttonPin = 2;
int ledPin = 13;
int buttonState = 0;

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(buttonPin, INPUT);
}

void loop() {
  buttonState = digitalRead(buttonPin);
  if (buttonState == HIGH) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
  }
}
```

- (a) Blinks an LED connected to pin 13 when a button connected to pin 2 is pressed
- (b) Turns on an LED connected to pin 2 when a button connected to pin 13 is pressed
- (c) Reads the state of a button connected to pin 2 and turns on an LED connected to pin 13 accordingly
- (d) Controls the speed of a motor based on the state of a button connected to pin 13
- (e) None of the above

42) What does the following Arduino code snippet do?

```
void setup() {
  pinMode(11, OUTPUT);
}

void loop() {
  analogWrite(11, 255);
}
```

- (a) Turn on an LED connected to pin 11 for 255 milliseconds
- (b) Control a servo motor connected to pin 11
- (c) Generate a square wave signal on pin 11
- (d) Turn on an LED connected to pin 11 with minimum brightness
- (e) Turn on an LED connected to pin 11 with maximum brightness

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Consider the following Arduino code snippet to answer the questions 43-45.

```
const int ledPin = 3;
int brightness = 0;
int fadeAmount = 5;

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  analogWrite(ledPin, brightness);
  brightness = brightness + fadeAmount;
  if (brightness <= 0 || brightness >= 255) {
    fadeAmount = -fadeAmount;
  }
  delay(30);
}
```

- 43) What does the provided Arduino code snippet accomplish?
- (a) Controls a servo motor's position based on an analog input signal
 - (b) Measures temperature using a temperature sensor and adjusts LED brightness
 - (c) Generates a PWM signal on pin 3 to control motor speed
 - (d) Flashes an LED connected to pin 3 with a fixed delay
 - (e) Fades an LED connected to pin 3 in and out with a smooth transition
- 44) What will happen when the brightness variable reaches either 0 or 255 in the provided Arduino code snippet?
- (a) The LED will stop fading
 - (b) The Arduino will enter a sleep mode
 - (c) The LED will blink rapidly
 - (d) The fade direction will reverse
 - (e) The Arduino will reset
- 45) Which communication protocol is commonly used for high-speed communication between Arduino boards and peripherals, allowing full-duplex data transfer?
- (a) Serial Peripheral Interface (SPI)
 - (b) Inter-Integrated Circuit (I2C)
 - (c) Universal Asynchronous Receiver-Transmitter (UART)
 - (d) Pulse Width Modulation (PWM)
 - (e) Analog-to-Digital Conversion (ADC)

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- 46) Which of the following components stores electrical charge and is commonly used to stabilize voltage in a circuit?
- (a) Resistor
 - (b) Inductor
 - (c) Transistor
 - (d) Capacitor
 - (e) Diode
- 47) Which of the following components is commonly used to connect electronic components on a breadboard?
- (a) Solder
 - (b) Jumper wires
 - (c) Resistors
 - (d) Capacitors
 - (e) Integrated circuits
- 48) What is the primary function of an operational amplifier (OP-amp) in electronic circuits?
- (a) To generate AC signals
 - (b) To amplify digital signals
 - (c) To amplify analog signals
 - (d) To convert analog signals to digital
 - (e) To perform logic operations
- 49) Which of the following best describes the ideal characteristics of an operational amplifier?
- (a) High input impedance, low output impedance, and infinite voltage gain
 - (b) Low input impedance, high output impedance, and finite voltage gain
 - (c) Low input impedance, low output impedance, and finite voltage gain
 - (d) High input impedance, low output impedance, and finite voltage gain
 - (e) High input impedance, high output impedance, and infinite voltage gain
- 50) Which type of battery is typically designed to provide a high discharge rate for short bursts of energy, such as in starting an engine?
- (a) Lithium-ion (Li-ion)
 - (b) Nickel-metal hydride (NiMH)
 - (c) Alkaline
 - (d) Lead-acid
 - (e) Zinc-carbon

1. The first part of the document is a list of the names of the persons who have been named in the proceedings.

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2. The second part of the document is a list of the names of the persons who have been named in the proceedings.